PAH-OMC-017: a common local Gibbs state on the fixed strip

TECT verification-first repository
claim `C6-SPACETIME-SIGNATURE`

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Purpose and scope

The single question is whether the R-509 fixed-graph cutoff-limit laws ν_n are Cauchy on each fixed finite-support bounded continuous amplitude cylinder as $n \to \infty$. The preregistration is `PAH-OMC-017-cauchy-prereg-v1.json`, with SHA-256:

`249bf12f71b4869e566925b8c011291ec7`
`4ef2fc4f5df2faeeafc4050f04fdff`

The immutable PAH-001 functional, OMC-004 strip, and OMC-016 state are unchanged. Use $K = 2$, $M_s = 1$, $\epsilon = 1/2$, $\beta = \nu = 1$, $m^2 = \theta = 0$, all other couplings one. First take $j \to \infty$ at fixed n on $R_j = 2^j$, $M_j = 2^{2j}$, then $n \to \infty$. The reference for ν_n is product dr times labelled counting. Zero-amplitude phase labels remain; all Q have the original weights $w_Q = Z_Q/Z$. No Jacobian, probe, gauge quotient, fitted prior, counterterm or altered rate is inserted. The terminal square and all boundary terms remain. The strip is relational: no physical dimension, length or volume is assigned. All constants below are analytic bounds, not fitted data.

Statement. On the frozen prefix Λ_m, consisting of columns $0, \ldots, m$, their vertical edges and the three cross links of every cell $i < m$, let $\mathcal{A}_m$ be the real bounded continuous gauge-invariant cylinder algebra. Anchor automorphisms are the declared identity. Comparison maps only forget new coordinates. There is a normalized positive local state ν_∞ and constants $0 < q < 1$, $D_m > 0$ such that

$$|\nu_n(f) - \nu_\infty(f)| \le 4\|f\|_\infty D_m q^{n-m} \quad \text{when } n \ge \max(2, m),\ D_m q^{n-m} \le \tfrac{1}{2}.$$

The constants are defined by the exact integral operator below; their existence is proved, but no certified numerical spectral gap is supplied. This is convergence of the specified static-state sequence, not a limiting dynamics construction or a theorem about all Gibbs boundaries.

Exact source-to-transfer dictionary

A column x has amplitudes $r_0, r_1 \ge 0$, apertures s_0, s_1, phase signs p_0, p_1 and vertical link sign v. Its measure ξ is $dr_0 dr_1$ times counting on 2^5 labels. Set

$$\begin{aligned} V(r,s) &= \tfrac{1}{2}(s-1)^2 + \tfrac{1}{6}r^6 + \tfrac{1}{4}r^4 + \tfrac{1}{2}s^2 r^2, \\ J(s,t) &= \frac{2}{s+t}, \\ E(x_a, y_b; U) &= \tfrac{1}{2}(s_a - t_b)^2 + \tfrac{1}{2}J(s_a, t_b)(p'_b r'_b - U p_a r_a)^2. \end{aligned}$$

Let $W(x) = V(x_0) + V(x_1) + E(x_0, x_1; v_x)$. For adjacent columns, $B_\triangle(x, y; h_0, h_1, d)$ is the sum of the two horizontal and one bottom-left to top-right diagonal edge energies, plus

$$\frac{J_{h_0} + J_{v_y} + J_d}{3}(1 - h_0 v_y d) + \frac{J_d + J_{h_1} + J_{v_x}}{3}(1 - d h_1 v_x).$$

The final $B_\square(x,y;h_0,h_1)$ has just the two horizontal energies and $\frac{1}{4}(J_{h_0}+J_{v_y}+J_{h_1}+J_{v_x})(1-h_0v_yh_1v_x)$. Thus every edge and each face retains its own original stiffness average.

Put $u(x)=e^{-W(x)/2}$ and define on $H=L^2(X,\xi)$

$$k_\eta(x,y)=u(x)e^{-B_\triangle(x,y;\eta)}u(y), \qquad K(x,y)=\sum_{\eta\in\{\pm1\}^3}k_\eta(x,y),$$

$$S(x,y)=u(x)\sum_{\theta\in\{\pm1\}^2}e^{-B_\square(x,y;\theta)}u(y), \qquad g=Su.$$

These are spatial partition operators, not e^{tL_ρ}. Every vertex/vertical edge occurs once in W, and every other edge and face once in its cell. For every configuration and every n,

$$F_n=\sum_{i=0}^{n+1}W(x_i)+\sum_{i=0}^{n-1}B_\triangle(x_i,x_{i+1};\eta_i)+B_\square(x_n,x_{n+1};\theta_n).$$

Interior half-onsite factors combine to one; outer u factors supply the two missing endpoint halves. The label count is exactly $5(n+2)+3n+2=2|V_n|+|E_n|=8n+12$. Tonelli gives

$$Z_n=\langle u,K^nSu\rangle=\langle u,K^ng\rangle.$$

In particular the unsplit square is in g, not deleted. This is a term/multiplicity identity, not assumed finite Gibbs projectivity.

Unbounded-domain and spectral estimates

All source terms are nonnegative, and $W\ge(r_0^6+r_1^6)/6$. Splitting the scalar integral at 2 gives $\int_0^\infty e^{-t^6/6}dt\le 3$, so $\|u\|_2^2\le 32\cdot 9$. Moreover $0<K(x,y)\le 8u(x)u(y)$ and $0<S(x,y)\le 4u(x)u(y)$. Thus both kernels are Hilbert–Schmidt and compact on the full unbounded space. Product-simple-function approximations give a finite-rank proof; no self-adjoint spectral expansion is needed. Strictly positive rows make K,K^*,S positivity improving, and $g\in H$ is strictly positive.

On the proof box $B=[0,1]^2$ times all labels, $\xi(B)=32$. The source bounds $V\le 25/24$, $E\le 33/8$, face energy ≤ 4 give $W\le 149/24$, $B_\triangle\le 163/8$ and $K(x,y)\ge 8e^{-319/12}$ on $B\times B$. With $h=1_B$, $Kh\ge 256e^{-319/12}h$. Hence

$$\lambda=r(K)\ge 256e^{-319/12}>0.$$

The box is a proof test function, not a cutoff or replacement state.

Applicable theorem and residual proof. The weak Krein–Rutman theorem applies: H is Banach, its positive cone is closed and total, and K is positive, compact with positive radius. It supplies positive eigenvectors $K\phi=\lambda\phi$, $K^*\psi=\lambda\psi$. Strict positivity follows from the kernel. Normalize $\|\phi\|_2=1$, $\langle\psi,\phi\rangle=1$. The solid-cone version does *not* apply to this L^2 cone.

If $Kw=zw$ and $|z|=\lambda$, then $K|w|\ge\lambda|w|$; pairing the difference with $\psi>0$ forces equality. Equality in the integral triangle inequality in a strictly positive row forces a single complex phase of w, hence $z=\lambda$. Two independent positive eigenvectors would yield a sign-changing real eigenvector by subtraction, impossible by the same argument. A generalized vector $(K-\lambda I)v=c\phi$ would give $0=c$ upon pairing with ψ. Thus the leading eigenvalue is algebraically simple and has no peripheral competitor. Compact spectral theory excludes accumulation at nonzero λ. For $Ph=\phi\langle\psi,h\rangle$ and $A=K/\lambda-P$, $P^2=P$, $PA=AP=0$ and $s=r(A)<1$.

Set $q=(1+s)/2$ and choose the least $p\ge 1$ with $\|A^p\|\le q^p$, which exists by the spectral-radius formula. Let

$$C=\max\{1,\|I-P\|,\max_{0\le r<p}\|A^r\|/q^r\}.$$

Writing $k = lp + r$ and using submultiplicativity proves $\|A^k\| \le Cq^k$. For $k \ge 1$, $\lambda^{-k}K^k = P + A^k$; for $k = 0$ the residual is $I - P$, separately covered. Therefore

$$\|\lambda^{-k}K^k - P\| \le Cq^k \quad (k \ge 0).$$

An independent derivation integrates $z^k(zI - A)^{-1}$ around $|z| = q$; the bounded resolvent gives the same exponential form. Neither derivation assumes $\|A\| = r(A)$, self-adjointness or fitted finite eigenvalues.

Decorated insertions and the Cauchy estimate

For $f \in \mathcal{A}_m$, integrate the left decorated prefix including f and $u(x_0)$, leaving x_m free; call the result w_f. Then $w_1 = (K^*)^m u > 0$ and $|w_f| \le \|f\|_\infty w_1$. Cross-link labels are summed *after* inserting f. Exactly, for $n \ge \max(2, m)$,

$$\nu_n(f) = \frac{\langle w_f, K^{n-m}g\rangle}{\langle w_1, K^{n-m}g\rangle}.$$

Let $d_m = \langle w_1, Pg\rangle > 0$, $a_f = \langle w_f, Pg\rangle$ and $D_m = C\|g\|_2\|w_1\|_2/d_m$. Positivity gives $|a_f| \le \|f\|d_m$. For $k = n - m$, numerator and denominator residuals are at most $\|f\|d_m D_m q^k$ and $d_m D_m q^k$. The latter is at most $d_m/2$ once $D_m q^k \le 1/2$. Applying

$$\frac{a+e}{d+\delta} - \frac{a}{d} = \frac{de - a\delta}{d(d+\delta)}$$

gives the stated bound with $\nu_\infty(f) = a_f/d_m$. For $F = \|f\| > 0$, put

$$N(\varepsilon, f) = \max\left\{2, m + \left\lceil \frac{\log\max(1, 2D_m, 8FD_m/\varepsilon)}{\log(1/q)} \right\rceil\right\}.$$

For $n, l \ge N$, $|\nu_n(f) - \nu_l(f)| \le \varepsilon$. The zero observable needs only $N = \max(2, m)$. The limiting prefix density is the decorated left weight times $\phi(x_m)$, normalized by $\langle w_1, \phi\rangle$. Integrating its last kernel using $K\phi = \lambda\phi$ proves neutral restriction consistency. This defines a positive normalized state on the common local algebra. The estimate applies to bounded measurable prefixes of ν_n, but not to discrete-to-continuous total variation in j.

Devil's-advocate audit and verification boundary

1. Missing half factors or the terminal square: rejected by the exact incidence bijection and hostile deletion controls.

2. A solid L^2 cone or symmetric kernel: those shortcuts are invalid and unused; weak eigenvectors plus strict-row and dual-pairing arguments give the required spectral conclusion.

3. Unbounded amplitudes or zero spectral radius: controlled separately by the Hilbert–Schmidt envelope and the positive proof-box test.

4. A vanishing normalization denominator: controlled by $d_m > 0$ and the explicit half-denominator threshold; that threshold cannot be dropped.

5. Static convergence implies dynamics or arbitrary-boundary uniqueness: upheld as objections to those stronger claims, which are excluded.

Primary and independent rational/symbolic implementations audit the same source energy and derived bounds. Hostile checks attack boundary factors, periodic positive matrices, nonnormal norms and normalization. Lean checks label/endpoint arithmetic and the actual normalized ratio inequality; the infinite-dimensional analytic and measure proof is not formalized. See the result manifest for exact fresh run counts and hashes. No external signed referee report is claimed; external review is invited.

The detailed applicability crosswalk is in `PAH-OMC-017-spectral-proof.md`. Imports are Lu–Lu (2022), Communications of the AMS 2, Theorem 5.2 (weak Krein–Rutman only); Schenker's MSU functional-analysis notes, Theorem 32.1; and Sorensen's LMU FA2 lecture of 14 November 2011 (spectral-radius formula). Their hypotheses are checked for this exact operator, not inferred from theorem names.

Result footer and next question

This closes only the preregistered static local-state Cauchy question. It supplies an auxiliary common-state input; T-054's active gate and C6 T1 remain unchanged. PAH-OMC-014 remains unresolved. No infinite-volume dynamics, full cutoff/continuum, physical Pre-A, spacetime, QFT, gravity, Yang–Mills, causal cone, mass gap or TOE follows. Next: freeze a separate domain and determine the j-limit of the original unaccelerated generators against this common state. A shrinking radial step may remove radial dynamics despite nondegenerate static amplitudes. No time rescaling is authorized by the present result.

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Result ID: R-510
Claim ID: C6-SPACETIME-SIGNATURE
Precise statement: The frozen nu_n converge on every
 fixed bounded-amplitude prefix, with the stated Cauchy bound.
Scope: Original OMC-004 strip and OMC-016 counting state;
 fixed K=2, M_s=1, epsilon=1/2, beta=nu=1; j before n.
Dependencies: Pinned PAH parents, R-509 and the audited
 weak Krein-Rutman and compact spectral theorems.
Evidence grade: ANALYTIC, EXACT, EXECUTED;
 inherited general spectral results, no physical promotion.
Reproduction command: python -X utf8
 verification/scripts/pah_omc017_verify.py --check
Expected output: All lanes PASS; diagnostics-free Lean.
Falsification gate: Missing source term, label, spectral
 hypothesis or positive-denominator/Cauchy estimate.
Tier before / after: T1 / T1 (host unchanged).
No-overclaim statement: No PAH-OMC-014 closure, limiting
 dynamics, full cutoff/continuum, physical Pre-A, spacetime,
 QFT, gravity, Yang-Mills, mass gap or TOE.
Next required action: Separate original-generator j-limit
 contract on a frozen bounded-amplitude domain.
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